Fiscal Year:	FY 2013	**Task Last Updated:**	FY 04/06/2013
PI Name:	Yokota, Hiroki Ph.D.		
Project Title:	Integrated Stress Response in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Organization Name:	Indiana University Purdue University Indianapolis		
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City:	Indianapolis	**State:**	IN
Zip Code:	46202	**Congressional District:**	7
Comments:	NOTE: PI moved back to Indiana University from Rensselaer Polytechnic Institute in January 2011 [Ed., 01/26/2011]. PI previously at Indiana University until August 2010 [Ed. 09/2010]		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	09/14/2010	**End Date:**	03/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle		
Flight Assignment:	STS-135		
Key Personnel Changes/Previous PI:	NOTE: Administrative PI is Eric Ledet, Rensselaer Polytechnic Institute (RPI). PI Yokota moved back to Indiana University Purdue University Indianapolis from RPI in January 2011. PI originally at Indiana University Purdue University Indianapolis and moved to RPI in August 2010. [Ed., 04/05/2011]		
COI Name (Institution):	Zhang, Ping (Indiana University) Ledet, Eric (ADMINISTRATIVE PI: Rensselaer Polytechnic Institute) Zhao, Liming M.D. (Indiana University)		
Grant/Contract No.:	NNX10AU51G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note: 4/05/2011] PI Yokota moved to Rensselaer Polytechnic Institute (RPI) in August 2010 and this grant (NNX10AU51G) was established at that time. PI has since returned to Indiana University Purdue University Indianapolis and the research is being conducted through a sub-contract from RPI (Administrative PI is Eric Ledet) to Indiana University Purdue University Indianapolis (PI is Hiroki Yokota). We address a set of questions: Does unloading evoke translational reprogramming through ISR-linked phosphorylation of eukaryotic initiation factor 2a (eIF2alpha)? If so, what mechanisms are responsible for selecting an ISR-linked pro-survival or apoptotic pathway? We hypothesize: • Transient unloading induces an ISR-linked pro-survival program by activation of eIF2alpha phosphorylation and reduction in translational efficiency. • Continuous unloading induces oxidative stress and apoptosis through accumulation of stress to the ER, but this chronic stress effect can be alleviated through a sustained phosphorylation level of eIF2alpha To test the above hypothesis and examine a potential role of ISR-driven signaling under unloading, we propose the following three specific aims using a mouse hindlimb suspension model: • Aim 1: Evaluate phosphorylation of eIF2alpha and its kinases (e.g., PERK and GCN2) as well as translational efficiency in response to transient unloading • Aim 2: Evaluate gene regulatory mechanisms linked to oxidative stress and stress to the ER in response to continuous unloading • Aim 3: Examine the effects of administration of salubrinal (inhibitor of eIF2alpha phosphatase) on pro-survival signaling and bone metabolism The project will employ real-time PCR, Western blot analysis, enzymatic assays, and bone histology with pharmacological agents, and eIF2a-linked knockout mice. Research schedule: The original timetable is unchanged. The first year will be focused on Aim1, in which the role of eIF2alpha phosphorylation, PERK/GCN2 kinases, and gene regulation in unloading will be investigated. In the second and the third years, Aims 2 and 3 will be conducted, respectively.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The result in the first year supports the hypothesis that unloading is a novel form of stress sources that induce integrated stress responses (ISR). This finding allows us to consider a novel therapeutic treatment that mitigates ISR under microgravity. The finding also contributes to our fundamental understanding about ISR on ground. In the second year, bone samples from STS-135 as well as bone samples from ground-based experiments were used. Both samples presented consistent results regarding loading and unloading effects. In the third year, the effects of administration of salubrinal on bone mineral density were evaluated using hindlimb suspended mice. The results clearly showed efficacy of salubrinal in suppressing unloading induced bone loss.
Task Progress:	Hindlimb unloading was conducted using the procedure previously described. C57BL6 female mice were anesthetized by inhalation of isoflurane, and the tail was inserted into a plastic tube of the tail harness and glued with super glue. The animal was placed head-down at approximately a 30 – 40° angle that prohibited the hindlimbs from reaching the ground. The tail harness was attached to a metal swivel (fishing wire) and connected to a metal support that horizontally spanned a housing cage from one corner to the other. The mouse was able to move along the horizontal metal support and access about 80% of the cage. Water (hydrogel placed on floor) and chow (placed on floor) were provided ad libitum in a plastic dish throughout the experiment. Bone samples (distal tibiae, and femur) were harvested at various time points during hindlimb unloading and after unloading release. We determined changes in BMD/BMC in the lumbar spine and femur, as well as body weight. Compared to the age-matched controls, unloaded mice significantly reduced their body weight. Administration of salubrinal partially suppressed unloading-induced loss of body weight. Compared to age-matched controls, unloading induced a significant loss of BMD (bone mineral density) and BMC (bone mineral content). Importantly, salubrinal suppressed reduction in BMD/BMC ($p<0.05$) compared to the unloading vehicle control. Salubrinal administration also suppressed unloading induced loss of BMD and BMC in the tibia. In consistent with an increase in BMD and BMC by salubrinal, differentiation of osteoblasts was also examined using colony forming unit (CFU) assays for osteoblasts (CFU-OBL) and fibroblast-like MSCs (CFU-F). The project contributed to a basic understanding of a novel pathway (eIF2 alpha mediated signaling) in bone remodeling. Collectively, the results clearly show that modulation of the phosphorylation level of eIF2a by administration of salubrinal stimulates new bone formation and suppresses unloading-induced bone loss. The question to be addressed in the next step is the effects of salubrinal on bone resorption driven by osteoclasts. Although data from this project indicates enhancement of bone formation by salubrinal, it is not clear whether salubrinal prevents bone resorption through inhibition of development of osteoclasts. Preliminary data indicate that salubrinal has a dual role – enhancement of bone formation and inhibition of bone resorption. We are interested in determining the role of salubrinal in osteoclastogenesis using CFU-M (macrophages), and examining gene regulation linked to development of osteoclasts. This study demonstrates that unloading-reduced bone mass is significantly restored by salubrinal. The result is consistent with salubrinal's stimulatory role on osteoblasts. The study provides evidence that salubrinal can be a potential therapeutic agent for preventing bone loss in disuse osteoporosis, which is encountered by astronauts under microgravity.
Bibliography Type:	Description: (Last Updated: 04/09/2013)

Articles in Peer-reviewed Journals	Zhao L, Tanjung N, Swarnkar G, Ledet E, Yokota H. "Regulation of eIF2a phosphorylation in hindlimb-unloaded and STS-135 space-flown mice." Adv Space Res. 2012 Sep 1;50(5):576-83. http://dx.doi.org/10.1016/j.asr.2012.05.024 , Sep-2012
Articles in Peer-reviewed Journals	Zhao L, Dodge T, Ledet E, Yokota H. "Knee loading stimulates bone formation in tail-suspended mouse hindlimb." Biol Sci Space. 2011 Dec;25(2-4):77-82. https://www.jstage.jst.go.jp/article/bss/25/2_4/25_112502010/_pdf , Dec-2011